

Trainings ▾

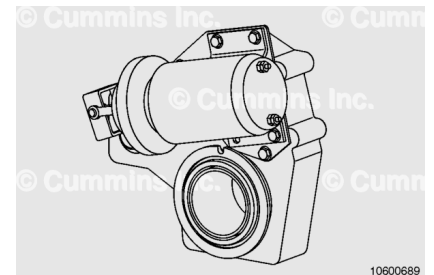
General Information

This air shutoff valve is normally open. For the valve to actuate, it **must** receive electrical power to operate the solenoid.

The air crossovers and valve assemblies for this engine contain keyways machined into the flange.

There is a pin attached to the v-band clamp to align the valve to allow for the best possible access.

Make sure to use the correct v-band clamp for the application.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



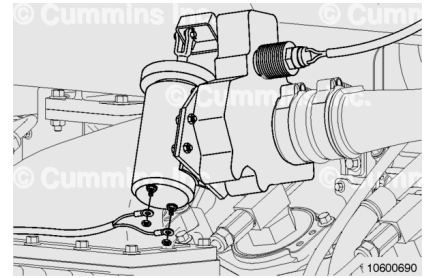
Note : This procedure is used for removing just ONE of the air shutoff valves from the engine. The remaining three can be removed using similar steps.

- Make sure the keyswitch is in the OFF position.
- Disconnect the batteries. Refer to Procedure 013-007 in Section 13. (</qs3/pubsys2/xml/en/procedures/56/56-013-007.html>)
- Remove the aftercooler heat shields. Refer to Procedure 010-129 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html>)

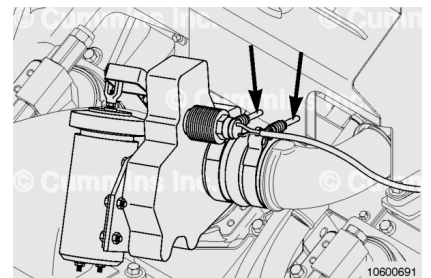
- Remove the turbocharger heat shields. Refer to Procedure 010-076 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html)
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Remove

Disconnect the electrical connector for the plunger switch and the ring terminal connections for the solenoid.



Remove the clamps on the hump hose (compressor) side of the valve.



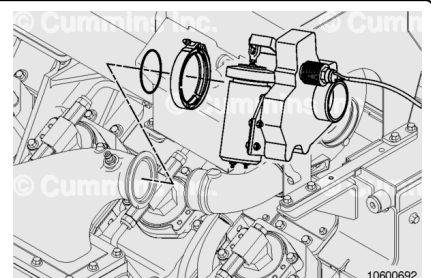
Support the valve assembly during removal.

Remove the V-band clamp from the air crossover connection.

Remove the valve.

Remove and discard the o-ring.

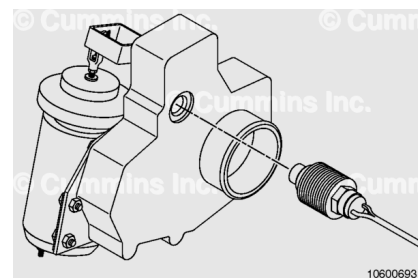
Cover the air crossover to prevent debris contamination.



Disassemble

Note : The valve body should not be disassembled because it is not a servicable part.

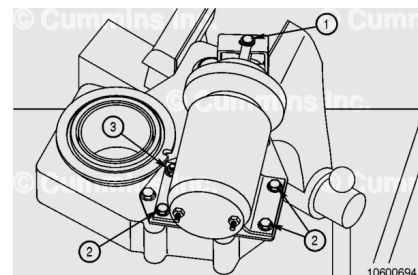
Remove the plunger switch from the housing. Use the following procedure in the Troubleshooting and Repair Manual, QSK38, QSK50, and QSK60 CM2150 Electronic Control System, Bulletin 4022102. Refer to Procedure 019-468 in Section 19. ([/qs3/pubsys2/xml/en/procedures/122/122-019-468.html](http://qs3/pubsys2/xml/en/procedures/122/122-019-468.html))



Note : Only remove the solenoid and mounting bracket if replacement of these parts or the valve body is required.

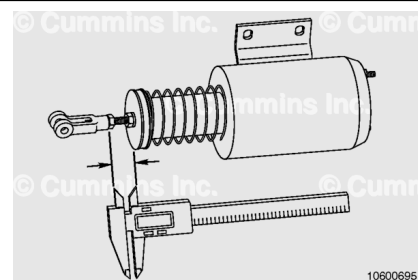
To remove the solenoid, secure the valve in a vise with the solenoid facing up.

Remove the old solenoid by loosening and removing the cotter key and the clevis pin (1), the three solenoid flange mounting capscrews (2), and the nut from the solenoid flange stud (3).

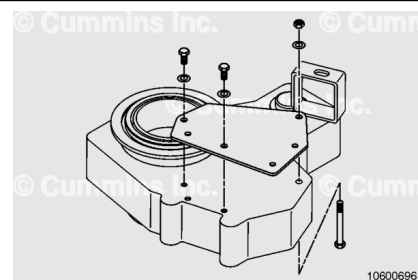


If replacing the solenoid, use calipers to measure the location of the solenoid end-clevis relative to the plunger face.

Record this value. It will be used as a reference during assembly of the valve to make the alignment process easier.



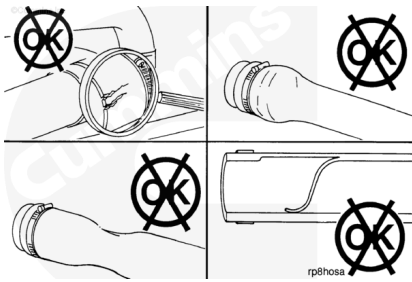
To remove the solenoid mounting bracket, remove the capscrews, nuts, and washers.



Clean and Inspect for Reuse

Inspect the hump hose and all clamps.

Replace any damaged parts.

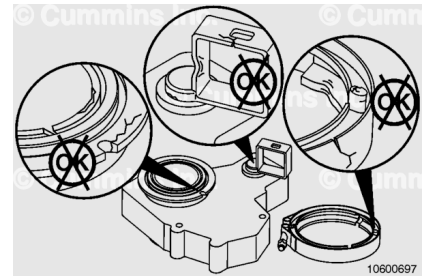


Inspect the v-band clamp to make sure the pin is intact.

Replace the clamp if any part of it is damaged.

Inspect the keyways on the air crossover and air shutoff valve body.

If either part is damaged, replace the damaged part.

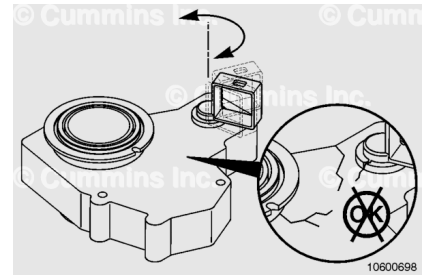


Inspect the valve body for cracks or other damage.

Rotate the linkage bar by hand to make sure that the valve moves freely. The valve body has a cast arrow showing the direction of operation from run (normally open) to stop (closed) positions.

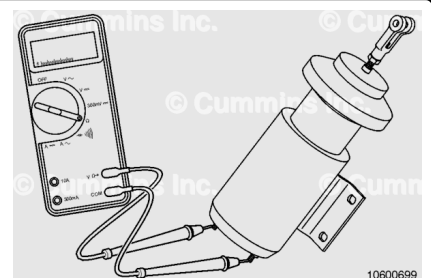
Replace the entire valve assembly if any damage is found or if the valve does **not** move freely.

If the seal element does **not** completely close, go to the Assemble section and check alignment.



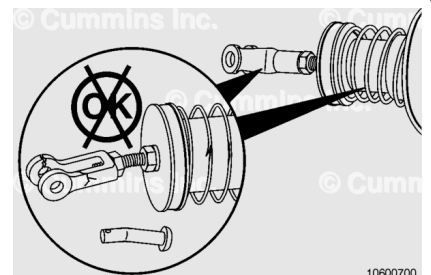
Inspect the solenoid by checking the resistance: 0.86 to 1.06 ohms.

Replace the solenoid if the resistance is outside of the specification.



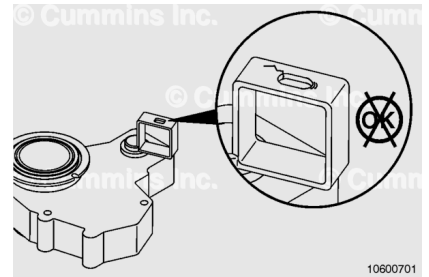
Inspect the clevis and spring, clevis pin, and solenoid shaft for damage.

Replace the solenoid if damage is seen.



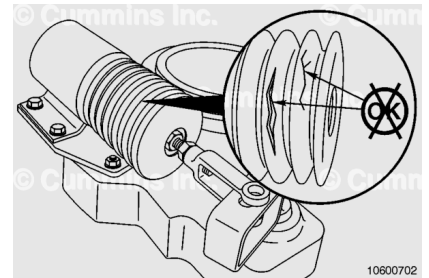
Check for damage to the slotted hole in the linkage bar.

If damage is found, the valve body **must** be replaced.



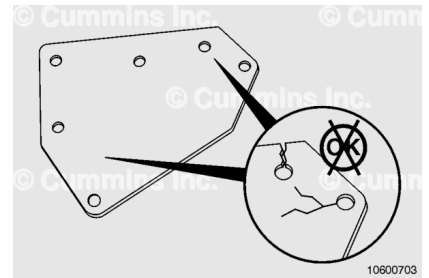
Inspect the rubber boot on the solenoid.

Replace the boot if torn, cracked, or otherwise damaged.



Inspect the solenoid mounting bracket for damage.

Replace the bracket if the mounting holes are worn or if the bracket is bent.



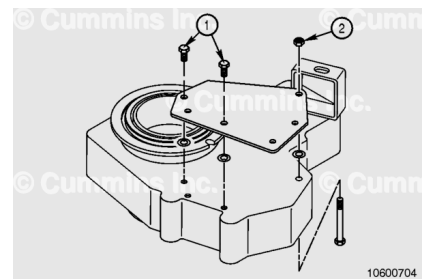
Assemble

The three washers should be placed under the solenoid mounting bracket.

Install the solenoid mounting bracket.

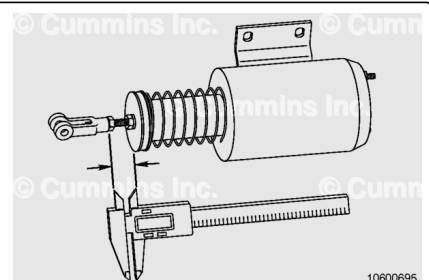
Tighten the bolts (1) and the nut (2).

Torque Value: 27 n•m [239 in-lb]



If reusing the clevis, use two wrenches to loosen and remove the jam nut and the solenoid end-clevis.

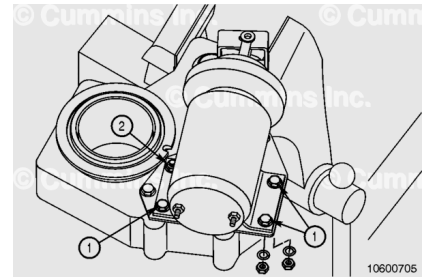
Thread the jam nut and clevis loosely onto the new solenoid, to approximately the same location as previously measured in the Disassemble section of this procedure.



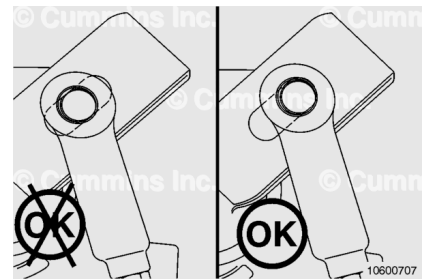
Place the solenoid onto the mounting plate. Make sure the clevis straddles the top of the linkage bar.

Insert the three bolts (1) with lock washers and nuts through the solenoid flange and steel mounting bracket and the lock washer and nut on the stud (2).

Torque Value: 24 n•m [212 in-lb]



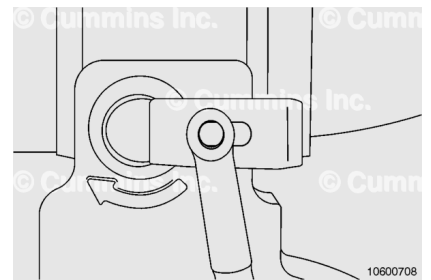
With the linkage bar in the open position, make sure that the clevis has sufficient travel.



Pull the plunger into the solenoid body and pull the linkage bar to the closed position.

The hole in the clevis should align with the round of the slotted hole as shown in the illustration to the right. None of the linkage bar should be visible through the hole in the clevis.

If the holes do not align, adjust the clevis in or out to make sure the holes line up in both the open and closed valve position.

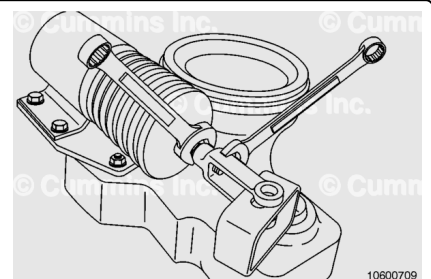


Lubricate the jam nut and clevis with high temperature anti-seize, Part Number 3824732.

Tighten the jam nut against the clevis.

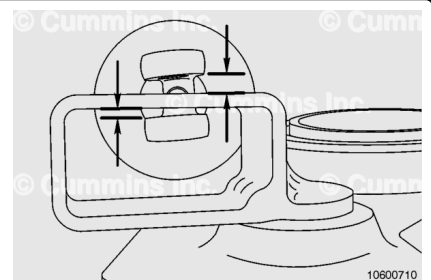
Torque Value: 8 n•m [71 in-lb]

Check that the jam nut is flush against the clevis. If there are any issues, check for damage to the jam nut or clevis and replace as necessary.



Pull the linkage bar to the closed position and check for clearance on both sides of the clevis.

Pull the linkage bar to the open position and check for clearance on both sides of the clevis.

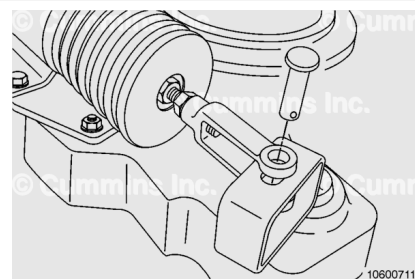


If either the clevis or plunger binds in either of these positions, the linkage bar and solenoid mounting plate should be rechecked for damage. If the issue persists, the valve body should be replaced.

Install the clevis pin **only** through the clevis and linkage bar.

Manually operate the valve to the closed position. Make sure that the valve is able to close completely. The circular indentation in the black-colored sealing element should be centered in the valve bore.

If the valve does **not** close completely, the Assemble section of this procedure should be repeated.



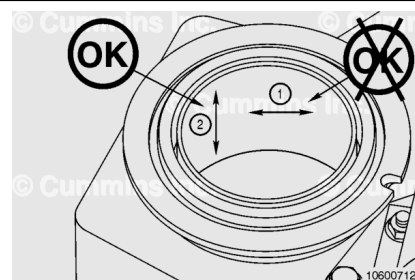
Hold the valve in the closed position using the linkage bar.

By hand, try to move the seal element in the direction of arrows (1). The seal element should **not** move.

If the seal element moves in the direction of arrows (1), adjust the clevis in or out.

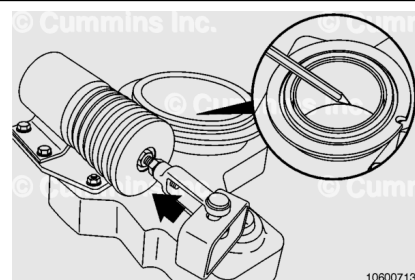
Move the seal element in the direction of the arrows (2). The seal element should be able to move.

If the seal element does **not** move in the direction of the arrows (2), adjust the clevis in or out.



Hold the valve closed by pushing the plunger into the solenoid as shown in the picture to the right. Do **not** use the linkage bar.

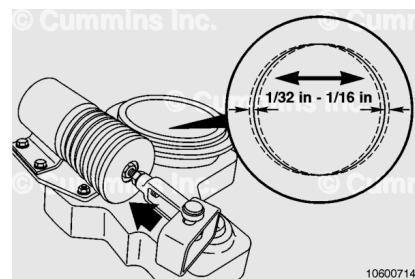
Make a pencil mark on the seal element where it meets the hub.



Hold the valve closed by pushing the plunger into the solenoid. Do **not** use the linkage bar.

By hand, try to move the seal element side to side. The pencil line should move approximately 1/32"-1/16". The seal element should also move up and down.

If the seal element movement is within this range, skip the next step.



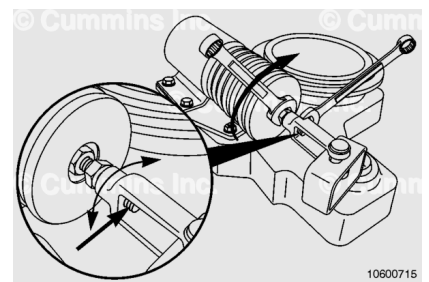
If the seal element movement is outside this range, complete this step.

Loosen the jam nut closer to the clevis.

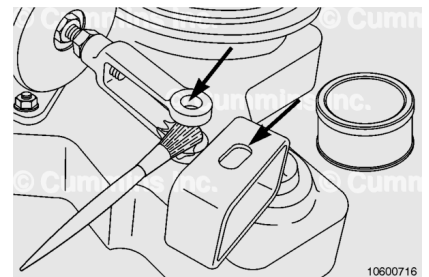
Rotate plunger shaft to allow approximately 1/32" – 1/16" movement of sealing element, then retighten jam nut into the clevis.

Torque Value: 8 n·m [71 in-lb]

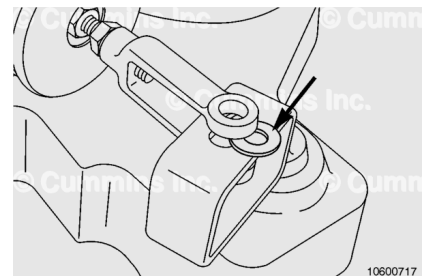
If the valve still does **not** seal properly, the Assemble section should be repeated.



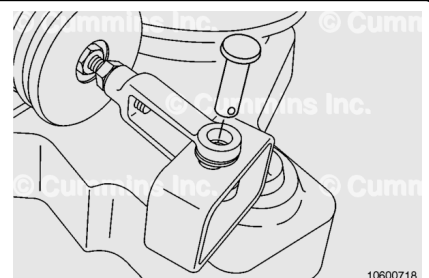
Grease the linkage bar slot and clevis with high temperature anti-seize, Part Number 3824732.



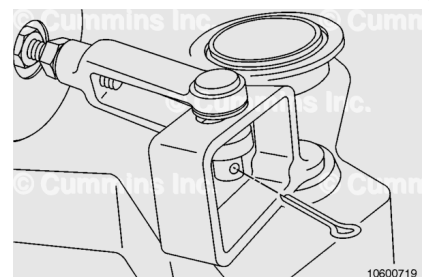
Place the tension washer between the top of the linkage bar and the clevis.



Insert the clevis pin through the clevis, washers, and linkage bar slot.

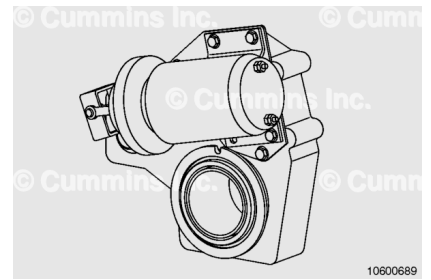


Put a cotter pin through the clevis pin. Wrap the cotter pin around the clevis pin.



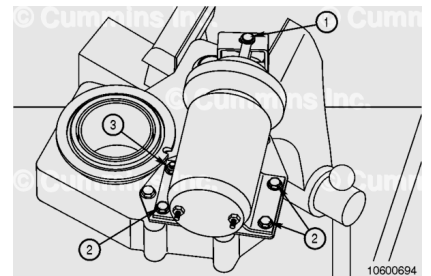
Open and close the valve, checking for free movement.

If the valve does **not** move freely, repeat the Assemble section of this procedure.



Install the plunger switch. Use the following procedure in the Troubleshooting and Repair Manual, QSK38, QSK50, and QSK60 CM2150 Electronic Control System, Bulletin 4022102. Refer to Procedure 019-468 in Section 19.

(/qs3/pubsys2/xml/en/procedures/122/122-019-468.html)



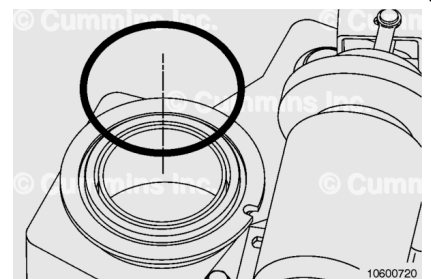
Install

Install a new o-ring on the valve assembly.

While supporting the valve, install the v-band clamp loosely on the air crossover connection closest to the aftercooler.

Both the valve assembly and the crossover are keyed.

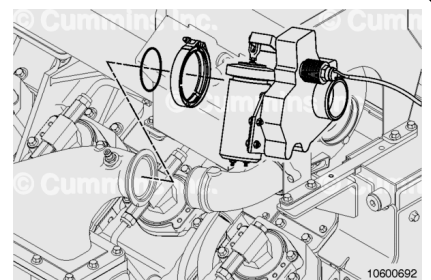
Note : The v-band clamp with the pin **must** be used. The pin orients the valve in the correct position. Without the pin, the valve will rotate and may interfere with the engine fuel lines.



Align the v-band clamp pin in the keyways.

Tighten the v-band clamp.

Torque Value: 11 n•m [97 in-lb]

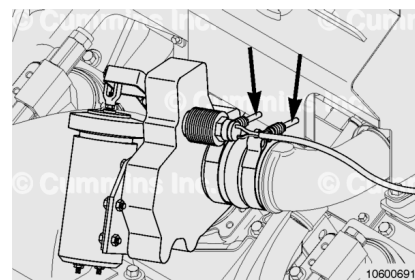


Loosely assemble the clamps on each end of the hump hose.

Place the hump hose on the straight connection closest to the compressor and slide it over the end of the valve assembly.

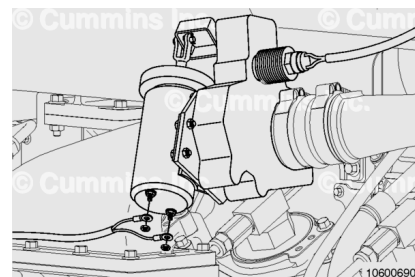
After checking proper alignment of the hose and clamps, tighten the clamps.

Torque Value: 11 n·m [97 in-lb]



Connect the connector for the plunger switch to the engine harness.

Connect the ring terminals to the solenoid.



⚠ CAUTION ⚠

Engine damage may occur if the Emergency Stop Button is pressed with the engine running.

If the solenoid has been replaced, the valve assembly should be tested using the following steps.

Connect the batteries. Refer to Procedure 013-007 in Section 13. (</qs3/pubsys2/xml/en/procedures/56/56-013-007.html>)

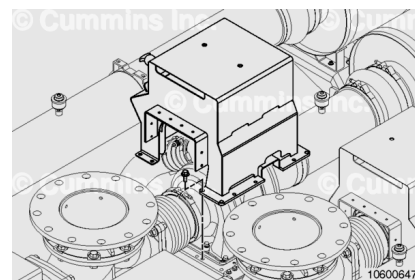
Turn the keyswitch to the ON position.

With the engine OFF, press the emergency stop button on the panel.

Check that the valve closes and returns open.

During normal operation, the valve will close and remain closed while the engine is pulling in air. The time the valve remains closed will depend on the operating conditions of the engine when the valve is closed.

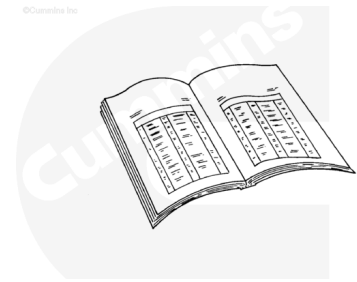
Note : If the valve is **not** working properly, repeat the Assemble, Inspect for Reuse, and Disassemble sections in this procedure



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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- Connect the turbocharger heat shields. Refer to Procedure 010-076 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html)
- Connect the aftercooler heat shields. Refer to Procedure 010-129 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html)
- Connect the batteries. Refer to Procedure 013-007 in Section 13. (/qs3/pubsys2/xml/en/procedures/56/56-013-007.html)
- Turn the keyswitch to the ON position.
- Operate the engine and check for leaks.